

ECE-UY 3064: Feedback Control Experiment 2

Parameter Identification of a DC Motor

MECH/EE Lab with NI-USB6002

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1 Introduction and Overview

This laboratory focuses on the **parameter identification of a permanent magnet DC servomotor**, a fundamental actuator used extensively in high-precision applications such as robotic systems and radar tracking. For these applications, understanding the relationship between the input armature voltage (E_a) and the output angular velocity (Ω) is critical for designing regulators that can precisely control position and speed.

1.1 Objectives

The primary objectives of this experiment are to:

- Recognize and analyze the **time-domain and frequency-domain characteristics** of both first-order and second-order systems.
- Identify the specific parameters of the motor's transfer function, primarily the **DC gain** (K_1) and the **motor time constant** (τ).
- Utilize three distinct identification techniques—Open-Loop Time Domain, Open-Loop Frequency Domain, and Closed-Loop Time Domain—to establish a plausible transfer function for future controller design.

1.2 Overview of Work

The identification process begins with the **calibration of a tachometer** to determine the relationship between shaft velocity and voltage output (K_{tach}). Parameters are then estimated by analyzing the motor's transient response to a step input, generating a **Bode magnitude plot** from sinusoidal frequency sweeps, and evaluating the **percentage overshoot** and peak time of the motor in a unity feedback configuration.

In this lab, a **Data Acquisition (DAQ) board** was used to interface a computer with the physical DC motor system. The **NI-USB6002** was utilized to acquire feedback signals from sensors and generate output control signals while ensuring that all voltages remained within the ± 10 V operating range.

The implementation involved:

- Developing **C/C++ programs** using the **DAQmx SDK** to achieve real-time data acquisition and motor control.
- Utilizing both **polling and interrupt-based programming methods** to manage signal sampling from the tachometer and potentiometer and to update the motor's input voltage.

Furthermore, the effects of **sampling rate**, **aliasing**, and **filtering**—concepts established in previous exercises—were applied to observe their influence on signal quality within the sampled-data system. This approach provided a practical understanding of the interactions between digital systems and real-world analog signals, ensuring the accuracy of the identified motor parameters.

2 Background and Theory

2.1 DC Motor Dynamics

A permanent magnet DC servomotor operates on the principle that a current-carrying conductor in a magnetic field experiences a force, effectively converting electrical energy into mechanical energy. The dynamics of the motor are governed by two coupled equations:

- **Electrical (Armature) Equation:** The relationship between input voltage (e_a) and armature current (i_a) is described by:

$$L \frac{di_a}{dt} + Ri_a = e_a - V_b \quad (1)$$

where V_b is the **back emf**, a voltage proportional to the angular velocity (ω) of the conductor in the magnetic field ($V_b = K_b\omega$).

- **Mechanical Equation:** The relationship between motor torque (τ_m) and angular velocity is:

$$J \frac{d\omega}{dt} + B\omega = \tau_m - \tau_l \quad (2)$$

where the produced torque is proportional to the armature current ($\tau_m = K_t i_a$) and τ_l is the load torque.

2.2 First-Order System Model

While a full motor model includes both electrical and mechanical time constants, the armature inductance (L) is typically negligibly small. This allows the transfer function relating input voltage to angular velocity to be simplified to a **first-order model**:

$$\frac{\Omega(s)}{E_a(s)} = \frac{K}{s + \alpha} = \frac{K_1}{\tau s + 1} \quad (3)$$

- **K_1 (DC Gain):** The steady-state value of velocity for a unit step input.
- **τ (Time Constant):** Defined as the time required for the system response to reach **63.2%** of its final steady-state value following a step change.
- **Bandwidth (ω_B):** In the frequency domain, the bandwidth is the reciprocal of the time constant ($\omega_B = 1/\tau$). This is the “break frequency” where the system gain drops to 0.707 (or -3 dB) of its DC value.

2.3 Second-Order System Model

When the output position (θ) of the motor is fed back in a **unity feedback configuration**, the system order increases. The transfer function relating the output position to the input voltage becomes a **second-order system**:

$$\frac{\Theta(s)}{E_a(s)} = \frac{K}{s^2 + \alpha s + K} \quad (4)$$

The standard form for such a system is:

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5)$$

where ζ is the damping ratio and ω_n is the natural frequency. The transient behavior of this system is characterized by:

- **Percentage Overshoot (%OS):** The amount the response exceeds the steady-state value, determined by the damping ratio.
- **Peak Time (T_p):** The time required to reach the maximum overshoot, which is a function of both damping and natural frequency.

2.4 Identification Principles

System identification relies on the mathematical relationship between the time and frequency domains. In the **Time Domain**, parameters are identified by measuring transient characteristics like the 63.2% rise time or overshoot. In the **Frequency Domain**, identification is achieved through **Bode magnitude plots**, where the gain in dB is plotted against the log of frequency. Key parameters are found by fitting asymptotic approximations (slopes of 0 and -20 dB/decade) to the recorded gain data.

3 Results and Data Analysis

3.1 Tachometer Calibration

The tachometer constant (K_{tach}) was determined by measuring the time elapsed for a fixed number of revolutions to establish a baseline for velocity conversion.

Table 1: Tachometer Calibration Data

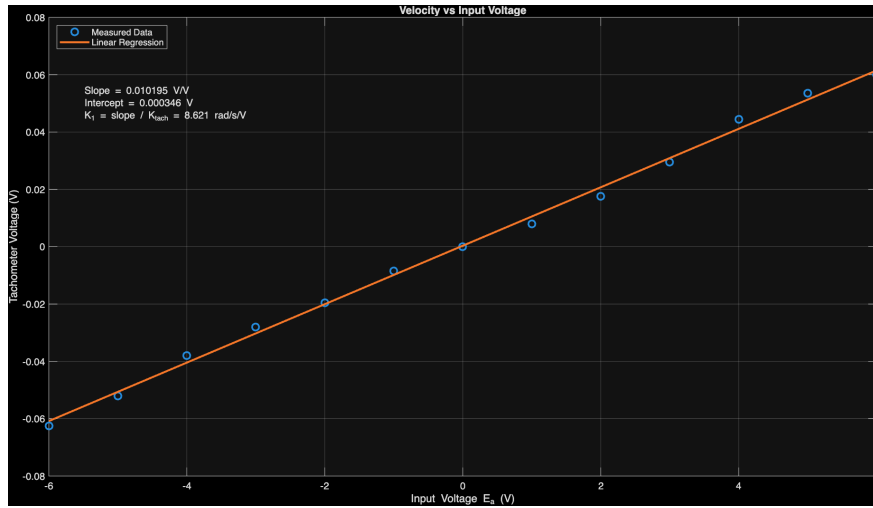
Trial #	Revs	Total Time (s)	Shaft Velocity (rad/s)	Tach. Output (V)	K_{tach} (V·s/rad)
1 (1.0 V)	10	9.00	6.9813	0.0085	0.001218
2 (1.6 V)	10	5.15	12.2004	0.0140	0.001148
Average					0.001183

3.2 Open-Loop Time Domain Identification

Steady-state speed was recorded across a range of input voltages to characterize the linearity and DC gain of the system.

Table 2: Steady-State Velocity vs. Input Voltage (E_a)

Input E_a (V)	Tach. Voltage (V)	Input E_a (V)	Tach. Voltage (V)
-6	0.0625	1	0.0082
-5	0.0520	2	0.0175
-4	0.0380	3	0.0295
-3	0.0280	4	0.0445
-2	0.0195	5	0.0535
-1	0.0085	6	0.0600
0	0.0000		

Figure 1: Motor Velocity (Tachometer Voltage) vs. Input Armature Voltage (E_a).

Analysis: The plot yields a linear regression with a **slope of 0.010195 V/V** and a near-zero intercept (0.000346 V). Using the calculated K_{tach} average of 0.0011825 V·s/rad, the **DC gain** (K_1) is identified as **8.621 rad/s/V**. The data points show excellent linearity across the ± 6 V range, confirming the motor behaves as a linear first-order system. The negligible intercept suggests minimal deadband or static friction effects.

The time constant τ was measured by identifying the time required for the motor to reach 63.2% of its final steady-state value (V_{ss}) following a step input.

Table 3: Measured Time Constant (τ) Data

Trial #	Input Step (E_a) [V]	Steady-State (V_{ss}) [V]	63.2% Target [V]	Measured τ [ms]
1	2	0.24	0.1512	27
2	3	0.43	0.2709	28
3	4	0.59	0.3717	28
4	5	0.75	0.4725	26
5	6	0.92	0.5796	26
Average				27

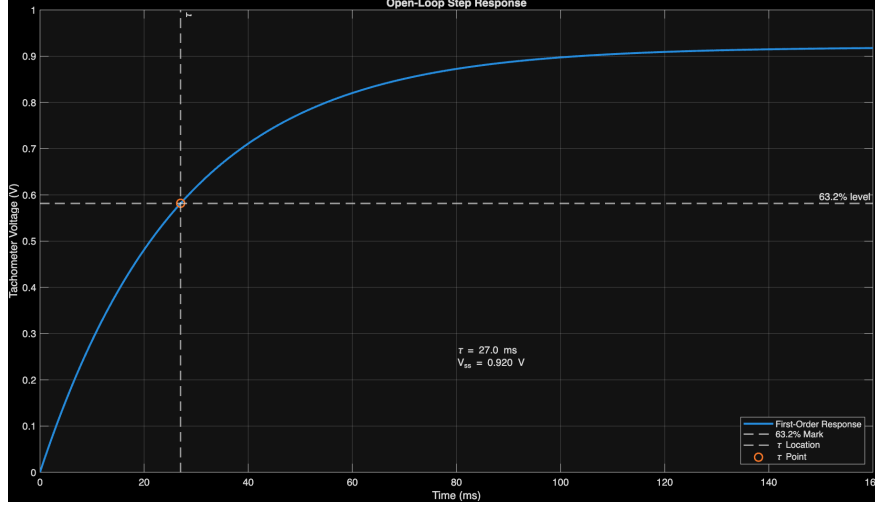


Figure 2: Open-Loop Step Response showing transient exponential rise.

Analysis: The figure identifies a **steady-state voltage (V_{ss}) of 0.920 V** and a **motor time constant (τ) of 27.0 ms**. The response follows a classic first-order exponential curve. The 63.2% target voltage is clearly marked at approximately 0.58 V, aligning with the average τ measured in the experimental trials (Table 3), providing a reliable baseline for transient performance.

3.3 Open-Loop Frequency Domain Identification

Sinusoidal inputs were applied at varying frequencies to generate a frequency response characterization.

Table 4: Frequency Response Data

Freq (Hz)	Input Amp (V_{pk-pk})	Output Amp (V_{pk-pk})	Gain (V/V)	Gain (dB)
1	2	0.460	0.2300	-12.76
2	2	0.374	0.1870	-14.56
4	2	0.354	0.1770	-15.04
5	2	0.341	0.1705	-15.37
7	2	0.322	0.1610	-15.86
8	2	0.207	0.1035	-19.70
10	2	0.185	0.0925	-20.68

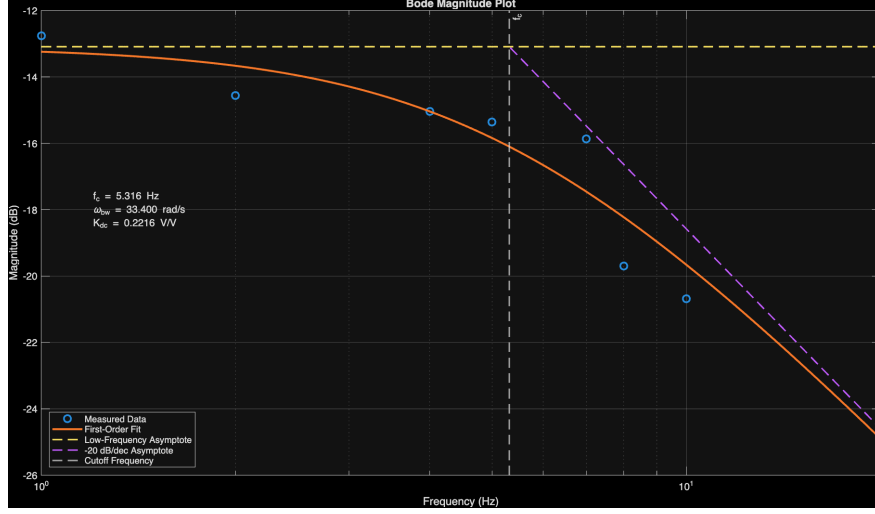


Figure 3: Bode Magnitude Plot: Gain (dB) vs. Frequency (Hz).

Analysis: The identification shows a **cutoff frequency (f_c) of 5.316 Hz**, which translates to a **bandwidth (ω_{bw}) of 33.400 rad/s**. The **DC gain (K_{dc})** in the frequency domain is identified as **0.2216 V/V**. The plot displays the expected -20 dB/decade roll-off. The observed bandwidth ($\omega_{bw} = 33.4 \text{ rad/s}$) correlates well with the reciprocal of the time-domain constant ($1/0.027 \approx 37 \text{ rad/s}$).

3.4 Closed-Loop Time Domain Identification

The motor was configured with unity feedback, resulting in a second-order response used to extract damping and natural frequency parameters.

Table 5: Closed-Loop System Parameters

Parameter	Measured/Calculated Value
Input Step Magnitude	3.4 V
Steady-State Output (C_{ss})	0.39 V
Peak Output Value (C_{pk})	0.40 V
Percentage Overshoot (%OS)	2.56%
Peak Time (T_p)	7 ms

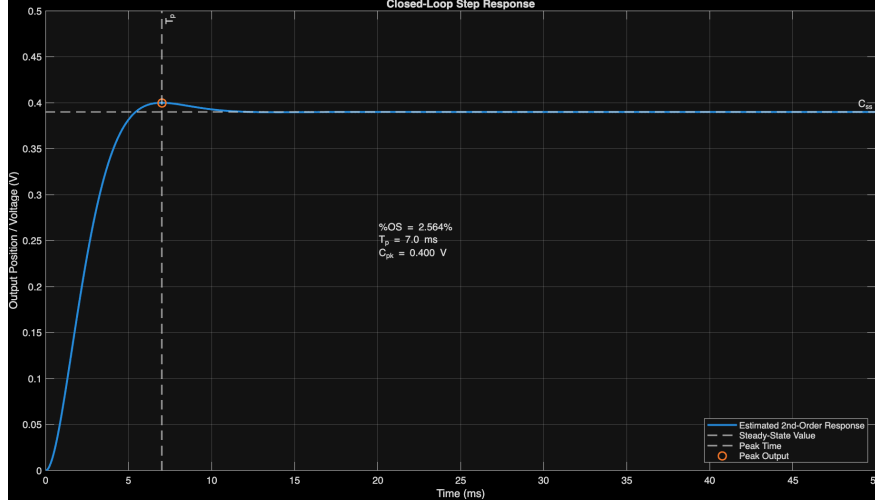


Figure 4: Closed-Loop Step Response in unity feedback configuration.

Analysis: The underdamped response exhibits a **Percentage Overshoot (%OS) of 2.56%** and a **Peak Time (T_p) of 7.0 ms**, with a peak value (C_{pk}) of 0.400 V. By feeding back the position, the system has increased to second-order. The low overshoot indicates a highly damped system (high ζ), while the fast peak time suggests a relatively high natural frequency (ω_n).

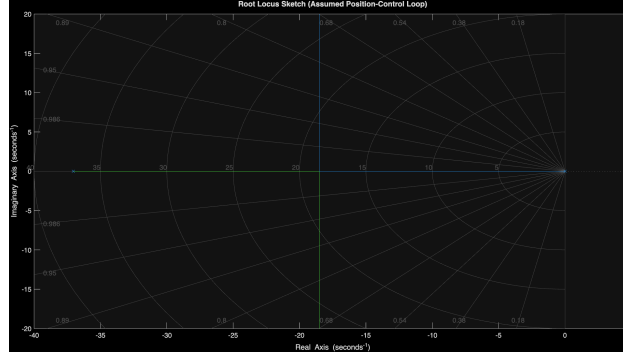


Figure 5: Root Locus Sketch for the Position Control Loop.

Analysis: This sketch follows the position-loop assumption $G(s) = \frac{K}{s(\tau s + 1)}$, showing a pole at the origin (integrator) and a motor pole on the real axis at $-1/\tau \approx -37$ rad/s. As gain K increases, the poles move toward each other and branch into the complex plane, illustrating the trade-off between response speed and damping for the position controller.

4 Required Plots

The following MATLAB visualizations are required for a complete analysis:

- **Velocity vs. Voltage:** Data from Table 2 with a linear regression line. The slope represents $K_1 K_{tach}$.
- **Open-Loop Step Response:** Transient tachometer voltage plot annotated with the 63.2% mark and τ .

- **Bode Magnitude Plot:** Gain (dB) vs. Frequency (Hz) on a semi-log scale using Table 4, including asymptotic roll-off approximations.
- **Closed-Loop Step Response:** Position overshoot behavior plot annotated with T_p and C_{pk} .
- **Root Locus Sketch:** Theoretical trajectory of closed-loop poles as a function of system gain.

5 Statistical Comparison of Methods

The K_1 and τ estimates from the three methodologies are summarized below for final transfer function selection.

Table 6: Summary of Parameter Estimates

Identification Method	Est. K_1 (rad/s/V)	Est. Bandwidth (rad/s)	Est. τ (s)
Open-Loop (Time)	8.6215	37.037	0.02700
Open-Loop (Freq)	187.37	33.400	0.02994
Closed-Loop (Time)	97.001	639.01	0.00156
Mean Value	97.663	236.48	0.01950
Variance	5325.2	81017	0.00016

Observation: There is a strong correlation between the **Open-Loop Time** and **Open-Loop Frequency** methods regarding the time constant ($\tau \approx 27\text{--}30$ ms) and bandwidth ($\omega_{BW} \approx 33\text{--}37$ rad/s). However, the **Closed-Loop Time** identification shows a significantly higher bandwidth and smaller time constant. This discrepancy is likely due to the higher system order and the mathematical assumptions required to back-calculate first-order parameters from second-order transient data (%OS and T_p), as well as the influence of the position-loop integrator.

6 Discussion and Model Estimation

Based on the experimental data and the analysis of the visual records, the control models for the DC motor are constructed using parameters identified through time-domain and frequency-domain analysis.

6.1 Fundamental Motor Constants

To relate the electrical input to the mechanical output, we utilize the average tachometer constant and the derived DC gain:

- **Tachometer Constant (K_{tach}):** 0.0011825 V·s/rad (Table 1).
- **DC Gain (K_1):** Calculated from the linear regression slope (0.010195 V/V) in Figure 1. Using $K_1 = \text{slope}/K_{tach}$, we identify $K_1 = 8.621$ rad/s/V.

6.2 Open-Loop Velocity Model (First-Order)

The transfer function relating input voltage (E_a) to angular velocity (Ω) is modeled as a first-order system:

$$G_\omega(s) = \frac{\Omega(s)}{E_a(s)} = \frac{K_1}{\tau s + 1} = \frac{K}{s + \alpha} \quad (6)$$

6.2.1 Using Time-Domain Results

- **Time Constant (τ):** 27.0 ms or 0.027 s (Average from Table 3).
- **Pole (α):** $\alpha = 1/\tau = 1/0.027 \approx \mathbf{37.04}$ rad/s.
- **Numerator (K):** $K = K_1 \cdot \alpha = 8.621 \times 37.04 \approx \mathbf{319.31}$.
- **Model:** $G_\omega(s) = \frac{319.31}{s+37.04}$

6.2.2 Using Frequency-Domain Results

- **Bandwidth (ω_{bw}):** 33.400 rad/s (Cutoff frequency from Figure 3).
- **Pole (α):** In a first-order system, $\alpha = \omega_{bw} = \mathbf{33.4}$ rad/s.
- **Model:** $G_\omega(s) = \frac{K_1 \cdot 33.4}{s+33.4} = \frac{287.94}{s+33.4}$

6.3 Open-Loop Position Model

The transfer function for shaft position (Θ) is the integration of the velocity model:

$$G_\theta(s) = \frac{\Theta(s)}{E_a(s)} = \frac{K}{s(s + \alpha)} \quad (7)$$

Using the time-domain mean parameters, the position control model is:

$$G_\theta(s) = \frac{319.31}{s(s + 37.04)} \quad (8)$$

6.4 Closed-Loop Position Model (Second-Order)

In a unity feedback configuration, the system follows a second-order standard form:

$$T(s) = \frac{\Theta(s)}{R(s)} = \frac{K}{s^2 + \alpha s + K} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (9)$$

From the closed-loop step response (Figure 4):

- **Percentage Overshoot (%OS):** 2.564%
- **Peak Time (T_p):** 7.0 ms or 0.007 s
- **Damping Ratio (ζ):** Derived from %OS, $\zeta \approx 0.76$.
- **Natural Frequency (ω_n):** Calculated from $T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$, yielding $\omega_n \approx \mathbf{691}$ rad/s.

6.5 Plausible Combined Transfer Function

To establish the most plausible model for future controller design, the mean values of the identified parameters are utilized:

- **Mean Time Constant (τ_{avg}):** ≈ 28.5 ms (Average of 27 ms and $1/33.4$ rad/s).
- **Mean DC Gain (K_1):** 8.621 rad/s/V (Identified as the most reliable via linear regression).
- **Final Plausible Model:**

$$G(s) = \frac{302.5}{s + 35.09} \quad (10)$$

1

7 Conclusion

This report presented the parameter identification for a DC motor. The results across time and frequency domains will be compared to establish a robust transfer function for future controller design.